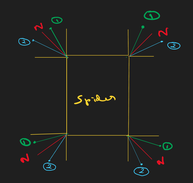
CRAWL GAIT OF QUAD

SPIDER ROBOT



$+x, +y$

FR Leg Front Right leg

① = [35.5, 72.0, -60.0] N + $\frac{\Delta y}{2}$

N = [35.5, 52.0, -60.0] neutral stance.

② = [35.5, 32.0, -60.0] ① − Δy

$-x, +y$

FL leg Front left leg

① = [-35.5, 72.0, -60.0] N + $\frac{\Delta y}{2}$

N = [-35.5, 52.0, -60.0] neutral stance.

② = [-35.5, 32.0, -60.0] ① − Δy

$-x, -y$

BL leg Back left leg

① = [-35.5, -72.0, -60.0] N + $\frac{\Delta y}{2}$

N = [-35.5, -52.0, -60.0] neutral stance.

② = [-35.5, -32.0, -60.0] ① − Δy

$+x, -y$

BR leg Back right leg

① = [35.5, -72.0, -60.0] N + $\frac{\Delta y}{2}$

N = [35.5, -52.0, -60.0] neutral stance.

② = [35.5, -32.0, -60.0] ① − Δy

FR → Front right
BL → Back left
FL = Front left
BR = Back right

gait cycle

① ② ③ ④
FR → BL → FL → BR

Phase : 1 — FR initiation

FR moves → N → ① → ② → N

FL, BL, BR moves from → N → ③ → N

Phase : 2 — initiation

BL moves → N → ① → ② → N

FL, BR, FR moves from → N → ③ → N

Phase : 3 — FL initiation

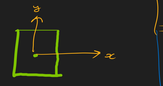
FL moves → N → ① → ② → N

FR, BR, BL moves from → N → ③ → N

Phase : 4 — BR initiation

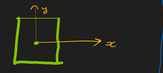
BR moves → N → ① → ② → N

FR, BL, FL moves from → N → ③ → N



movement in $\left(\frac{\Delta y}{4}\right)$

Stride length ÷ 4



movement in $\left(\frac{\Delta y}{4}\right)$

Stride length ÷ 4

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Stride length ÷ 4

COG movement
$\left[0, \frac{\Delta y}{4}\right]$
$\left[0, \frac{\Delta y}{4} \times 2\right]$
$\left[0, \frac{\Delta y}{4} \times 3\right]$
$\left[0, \frac{\Delta y}{4} \times 4\right]$

So AFTER full phase :

① → ② → ③ → ④

So AFTER full cycle :

So COG will be : moved by $[0, \Delta y]$: